

Generalized Self-Similar Solutions of Subsonic Marshak Waves

Theory and Numerical Validation Structure

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Inertial Confinement Fusion (ICF)

- Hohlraum Modeling Crucial for radiation temperature prediction and experimental measurement accuracy.
- Energy Delivery Key to analyzing laser–walls coupling and refining pulse shaping techniques.
- Historical Context Builds upon classical foundations established by *Lindl (1995)*, *Kauffman (1995)*, and *Rosen (2005)*.

Adding Shay & Menahem's suggestions

- Slides about the Aluminum article...

The Theory Of Subsonic Heat Waves

Physics Of The Problem

- We treat a semi-infinite wall subjected to high-temperature boundary heating.
- Dynamic equilibrium is mediated strictly via **photon transport**.
- In the **optically-thick limit**, an exceptionally sharp temperature front develops.

The Transfer Equation Coupled to Euler Equations

- Coupled Boltzmann & internal energy equation.
- The diffusion equation under the gray approximation.
- Euler equations for the hydrodynamic evolution of the material - in two words.
- The EOS for different materials fitted by H & R (2003 or previously).

The Supersonic Regime

The Equations of the Supersonic Regime

- "Black Physics" - Using the LTE assumption.
- The image of the supersonic regime from Shussman.
- Discussing the equivalence to $\chi \rightarrow \infty$.

The Subsonic Regime

The Equations of the Subsonic Regime

- Present the Euler equations for inviscid & incompressible (adiabatic) fluid flow.
- The image of the subsonic regime from Shussman.
- Explain about the assumed EOS of
- Discussing the equivalence to $\chi \rightarrow \infty$.

Some recent works in the field of subsonic wave

- Shussman-Heizler (2015) - a full solver using the patching method (More on next slide).
- Pakula & Sigel (1985) - the strong shock approximation.
- Hammer & Rosen (2003) - New solutions based on perturbation theory.
- Granier (2006) - solved the full domain for a specific BC.

The patching method

- Elaborating on the problem solved (see Shussman presentation).
- The assumptions made - strong shock approximation, vanishing ρ_0 and diverging ρ_f .
- Discuss the Hugoniot relations and usefulness for the solver method (**how much and whether I should talk about the solver's logic?**)
- The assembling of two different self similar problems.
- Solving for the full domain using a special patching method.

Self similarity (Optional)

Diving into self similarity (emphasizing the usefulness of the method)

- Taking the Pi-Backingham theorem slide from Menahem.
- Showing the unit-analysis table for both regions.
- system of PDEs $\implies$ system of ODEs (yielding a semi-analytical solution).

The actual research

What We Bring To The Table

- Value - Building a *benchmark* for IMC simulation codes (**Shay remarks that Steinberg has already debugged his IMC simulation with the solution from our code**).
- Generality - different materials, varying density at the shock domain, different cases, Eulerian coordinates.
- Method - fitting Shussman-Heizler's self-similar solutions to provide precise solutions for profiles and other important values produced by the solvers **in all times**.
- Practical - provide our work to your favorite AI agent and let it test itself in the subsonic regime.

Example Of The Usefulness Of Fittings

Giving an example for a profile & the total energy in the domain

- Presenting the photo of, for example, the ablative pressure profile (looks good and relatively simple but representative of some other fits).
- Presenting the total energy integral curve - with the expression in the ablation region that is explicit in the language of the hydro profiles.
- Describing the 3 options of our users when they get our results - either use a formula, or a table,

Bridging the BCs between IMC to solver/simulation

Marshak Boundary Condition (IMC)

- Explain in one or two words the general idea (Shay said the audience will be familiar with the theory).
- Present the equation that includes T_s (surface temperature) and T_{bath} (the effective temperature produced by the outside environment).
- Present the graph of $T_s(t)$ vs $T_{bath}(t)$ for the base case of $\tau = 0$.

T_{Bath} vs $T_{surface}$

- Which accepts what (IMC accepts T_{bath} , solver & simulation work with T_s).
- Deriving a self similar equation for T_{bath} for any $T_s(t) = T_0 t^\tau$.

NOTE

I should understand where it should be included, it's important as it bridges over the gap between IMC and the solver or simulation in terms of BC. **Maybe I should combine the "example of our work" & "marshak BC" slides into the "simulation setup" slide as a general example of the solutions**

The actual research

UNEDITED Simulation Setup: Case 1 ($\tau = 0$)

Medium Framework: Gold Target

The numerical validation uses an ideal gas with polytropic index $\gamma = \frac{5}{4}$ and initial high material density $\rho_0 = 19.32 \text{ g/cm}^3$ [cite: 477, 480].

Specific Internal Energy: [cite: 481]

$$e(T, \rho) = 3.4 \times 10^{13} \left(\frac{T}{\text{HeV}} \right)^{1.6} \left(\frac{\rho}{\text{g/cm}^3} \right)^{-0.14} \frac{\text{erg}}{\text{g}}$$

Rosseland Opacity Coefficient: [cite: 481]

$$\kappa_R(T, \rho) = 7200 \left(\frac{T}{\text{HeV}} \right)^{-1.5} \left(\frac{\rho}{\text{g/cm}^3} \right)^{0.2}$$

Boundary Conditions & Self-Similar Evolution ($\tau = 0$)

Surface drive is maintained via a fixed thermal profile $T_s(t) = 1 \text{ HeV}$ [cite: 479, 485]. The dynamic radiation bath drive and the calculated mass front position trace as [cite: 488, 489]:

$$T_{\text{bath}}(t) = \left[1 + 0.169141 \left(\frac{t}{\text{ns}} \right)^{-\frac{79}{192}} \right]^{1/4} T_s(t)$$

fits

- 4 slides - T , P , ρ , u .
- Touch in two words about chosen more complex fitting strategies that involved clever choice of fitting expressions or uses of the EOS in different domains.
- Slides for verification of the use of the profile fits in dimensional plots.
- Show slides with Eulerian coordinates in dimensional plots.
- Show slides with the Eulerian variables and plot them as axvline (the fronts for example) on the dimensional plots.

Presentation Summary & Plans for ahead

- A complete benchmark for IMC simulations in a variety of test cases - different materials, varying density.
- User-friendly, your favorite agent will love that, and it is guranteed to save you a valuable amount of time in debugging and especially authenticating your simulation against a subsonic solver (in the high-opcaity regime) which is backed-up by a well-established theory (Shussman & Heizler, 2015).
- Plans for ahead - fittings for the problem in gray-physics, improving the solver with different EOS, applying more complex multi-group approximations.

Archived Shussman slides

Our Work – Overview & Strategy

- ****Decoupled Patching:**** Instead of a single complex hydro-transport fit, we solve the ablation zone and the shock zone independently, later matching them via self-consistent boundary logic[cite: 69, 471].
- ****Self-Similarity:**** In each localized frame, we systematically neglect one governing dimensional parameter to yield exact self-similar ODEs[cite: 70].
- ****Inter-Zone Coupling:**** The thermodynamic properties calculated at the heat front serve directly as the driving boundary condition for the downstream shock equations[cite: 71].

Parameter Pool	Ablation Region	Shock Region
Density Profile	Assumed $\rho \rightarrow 0$ at source [cite: 95, 98]	Captures background
Radiation Flux (B)	Fully Coupled [cite: 90, 94]	Neglected ($B \approx 0$)

Table: Strategic scaling assumptions across regions.

Statement of the Problem

The 1D radiation-hydrodynamic transport framework in Lagrangian mass coordinates m is stated as[cite: 86]:

$$\begin{aligned}\frac{\partial V}{\partial t} &= \frac{\partial u}{\partial m} \\ \frac{\partial u}{\partial t} &= -\frac{\partial P}{\partial m} \\ \frac{\partial e}{\partial t} + P \frac{\partial V}{\partial t} &= \frac{\partial}{\partial m} \left(\frac{c}{3\kappa_R} \frac{\partial (aT^4)}{\partial m} \right)\end{aligned}$$

Combined Nonlinear Energy Equation

By tracking power-law material properties, the energy diffusion scales as[cite: 90]:

$$\frac{1}{r} \frac{\partial PV}{\partial t} + P \frac{\partial V}{\partial t} = B \frac{\partial}{\partial m} \left(V^\lambda \frac{\partial}{\partial m} (PV^{1-\mu})^{\frac{4+\alpha}{\beta}} \right)$$

where the structural constant is $B = \frac{16\sigma}{3(4+\alpha)} g(rf)^{\frac{-(4+\alpha)}{\beta}}$ [cite: 90].

Thermodynamic & Material Modeling

To construct a closed analytic framework, variables are mapped using power-law dependencies[cite: 73]:

Boundary Heating Profile

$$T(t) = T_0 t^\tau$$

Specific Internal Energy

$$e = f T^\beta \rho^{-\mu}$$

Rosseland Mean Opacity

$$\frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda}$$

Equation of State (EOS)

$$P = r \rho e \equiv (\gamma - 1) \rho e$$

Background density is initialized uniformly to ρ_0 ahead of the wavefront[cite: 78].

The Ablation Region Properties

The ablation zone is completely characterized by four dimensional parameters: B , T_0 , $\rho_0(V_0)$, and t [cite: 94]. Transforming to the dimensionless similarity coordinate ξ scales the structural variables[cite: 101]:

- ****Lagrangian Coordinate Scaling:**** [cite: 101]

$$m = \xi \cdot \left(B^{2-\mu} T_0^{8+2\alpha-2\beta+\beta\lambda-(4+\alpha)\mu} t^{2+2(4+\alpha-\beta)\tau-\mu(3+(4+\alpha)\tau)+\lambda(2+\beta\tau)} \right)^{\frac{1}{4+2\lambda-4\mu}}$$

- ****Local Structural Functions:**** Evaluated using dimensional scaling tracks [cite: 102]

$$X(m, t) = \tilde{X}(\xi) B^{w_{X1}} T_0^{w_{X2}} t^{w_{X3}}$$

Boundary Conditions for Ordinary Differential Equations

Boundary bounds are enforced exactly at the zero-volume ablation edge ξ_F [cite: 106]:

$$\tilde{V}(\xi_F) = 0, \quad \left. \frac{\partial \tilde{V}}{\partial \xi} \right|_{\xi_F} = 0, \quad \tilde{P}(0) = 0, \quad \tilde{u}(\xi_F) = 0$$

The Shock Region Integration

Once the ablation scale maps the pressure at the backend boundary, the shock zone is localized via $P(t) = P_0 t^{\tau_S}$ [cite: 139]. Neglecting active radiation transport ($B \rightarrow 0$) yields classic strong shock relations [cite: 140, 141]:

Rankine-Hugoniot Mass, Momentum, and Energy Conditions

$$\frac{D - u_S}{V_S} = \frac{D}{V_0}, \quad P_S = \frac{Du_S}{V_0}, \quad P_S u_S V_0 = D \left(e_S + \frac{u_S^2}{2} \right)$$

- ****EOS Divergence:**** To retain generality, the material model inside the shock domain is permitted to utilize alternative polytropic definitions ($r \neq r'$) [cite: 144, 146].
- ****Shock Coordinate Mapping:**** mass variables scale via [cite: 180]:

$$m_S = \xi P_0^{1/2} V_0^{-1/2} t^{1 + \frac{\tau_S}{2}}$$